Groups acting over vector spaces

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We will be working over $\mathbf{Vect}_k$ for some arbitrary field k of characteristic 0.

Note that we will only be considering actions of finite groups.

Given a finite group G acting on a k-Vector space V by the action

$$\lambda \in \mathrm{Hom}_{\mathbf{Grp}}(G, \mathrm{Aut}_{k\text{-Vect}}(V)),$$

we can define **λ-invariant maps** from $V \to W$ as those k-linear maps which respect the action of λ; i.e.,

$$\begin{aligned} \lambda\text{-invar}(V,W) &\subseteq \mathrm{Hom}_k(V,W) \\ \forall \varphi \in \mathrm{Hom}_k(V,W), \varphi \in \lambda\text{-invar}(V,W) &\Leftrightarrow [\forall v \in V, \forall g \in G, \varphi(v) = \varphi(\lambda(g)(v))] \\ &\Leftrightarrow [\forall g \in G, \varphi = \varphi \circ \lambda(g)]. \end{aligned}$$

When we refer to λ-invariant maps without first defining the action λ, λ is supposed to be inferred from context.

Now is a good time for some examples.

- Consider the symmetry group $S_2 = \{\mathrm{id}, \sigma\}$ acting on $V \otimes V$ by $\sigma(v_1 \otimes v_2) = v_2 \otimes v_1$. The λ-invariant maps from $V \otimes V$ to W are just symmetric k-linear maps from $V \otimes V$ to W, which are defined as symmetric bilinear maps from $V \times V$ to W.
- More generally, if we take the action of S_n on $T^n(V)$ as

  $$\sigma(v_1 \otimes \ldots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \ldots \otimes v_{\sigma^{-1}(n)},$$

  then we get symmetric multilinear maps.
- If we instead consider S_n acting on $T^n(V)$ by

  $$\sigma(v_1 \otimes \ldots \otimes v_n) = \mathrm{sgn}(\sigma) \cdot v_{\sigma^{-1}(1)} \otimes \ldots \otimes v_{\sigma^{-1}(n)},$$

  then we get *alternating* multilinear maps.

Now, we would be very happy if we could think of these λ-invariant maps V to W as just k-linear maps from some vector space to W. That is, we would like to have a vector space $X_{V,\lambda}$ such that there is a "natural" isomorphism between

$$\lambda\text{-invar}(V,W) \cong \mathrm{Hom}_k\big(X_{V,\lambda}, W\big). \tag{$*$}$$

In fact, if $(V,W) \mapsto \lambda\text{-invar}(V,W)$ and $V \mapsto X_{V,\lambda}$ are functorial, then we can formalize naturality by saying that the isomorphism is natural in parameters V and W.

Define V_λ as

$$V_\lambda := \frac{V}{\operatorname{span}_k\left(\bigcup_{\sigma\in G}\{v-\sigma(v) : v\in V\}\right)} = \frac{V}{\operatorname{span}_k(\{v-w : v\in V, w\in \operatorname{orb}_\lambda(v)\})}.$$

The universal property of the quotient tells us that V_λ satisfies the universal property described for $X_{V,\lambda}$ at $(*)$.

Suppose we have a subspace $W \subseteq V$, and we would like to find a subspace of V which is isomorphic to V/W. If we can find an endomorphism $f \in \operatorname{End}_k(V)$, such that $\ker(f) = W$, then we have, by the first isomorphism theorem,

$$\frac{V}{W} = \frac{V}{\ker(f)} \cong \operatorname{im}(f) \subseteq V$$

Consider Reynold's operator $f_\lambda \in \operatorname{End}_k(V)$, which is defined as

$$\forall v \in V, f_\lambda(v) := \frac{1}{|G|}\sum_{\sigma\in G}\sigma(v)$$

Observe that $\ker(f_\lambda) = \operatorname{span}_k(\{v-w : v\in V, w\in \operatorname{orb}_\lambda(v)\})$, and therefore, we get that

$$V_\lambda = \frac{V}{\ker(f_\lambda)} \cong \operatorname{im}(f_\lambda) \subseteq V$$

Let us call $\operatorname{im}(f_\lambda)$ as V^λ.

So, we have successfully identified V_λ, which is defined as a quotient space of V, with the subspace $V^\lambda \subseteq V$.

Recall that a linear operator $\varphi \in \operatorname{End}_k(V)$ is called projection iff $\varphi \circ \varphi = \varphi$.

One of the nice things about projection maps is that

$$\operatorname{im}(\varphi) = \operatorname{fix}(\varphi),$$

where $\operatorname{fix}(\varphi) = \{v \in V : v = \varphi(v)\}$ is the set of fixed points of V under φ.

Furthermore, f_λ is a projection map, which gives us two nice characterizations of V^λ,

$$V^\lambda = \operatorname{im}(f_\lambda) = \operatorname{fix}(f_\lambda) = \{v\in V : \operatorname{orb}_\lambda(v) = \{v\}\} = \{v\in V : \forall \sigma\in G, \sigma(v) = v\}$$

$$V^\lambda = \operatorname{im}(f_\lambda) = \left\{\frac{1}{|G|}\sum_{\sigma\in G}\sigma(v) : v\in V\right\}$$

In the case of symmetric multilinear maps, the vectors in $T^n(V)^\lambda \subseteq T^n(V)$ are known as the symmetric tensors. V_λ is denoted as $\operatorname{Sym}_n(V)$.

If V has a basis $\{v_1, ..., v_d\}$, then $\left\{v_{i_1} \otimes ... \otimes v_{i_n} : 1 \leq i_1 \leq ... \leq i_n \leq d\right\}$ is a basis for V_λ (*not* for the subspace V^λ) . So

$$\dim(\mathrm{Sym}_n(V)) = \binom{\dim(V) + n - 1}{n}.$$

In the alternating multilinear case, the vectors in $T^n(V)^\lambda$ are known as alternating tensors. V_λ is denoted as $\Lambda^n V$, and is known as the nth exterior product or the nth wedge product. f_λ is denoted as Alt^n.